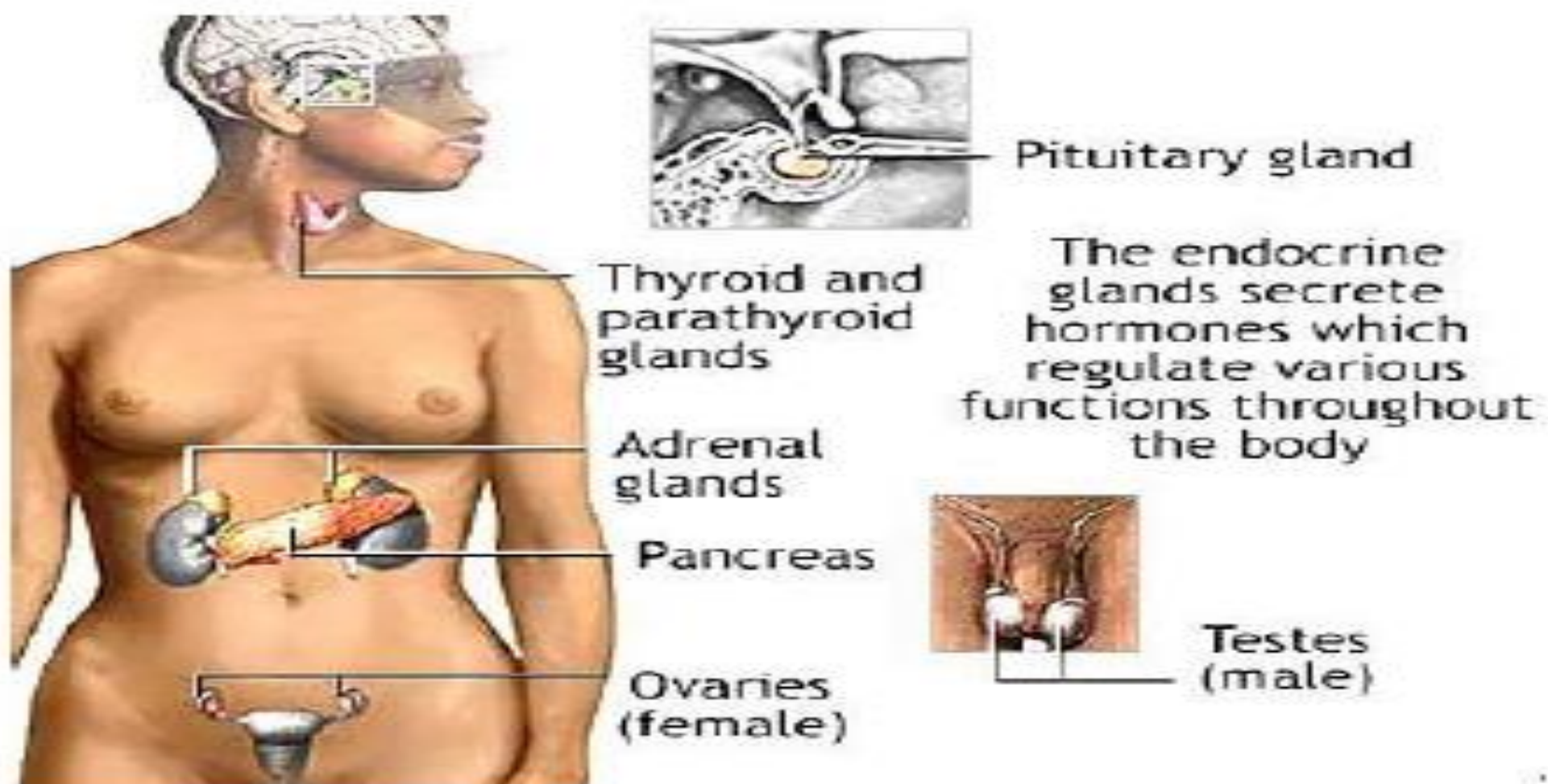


Anatomy of Nervous system(NS)

Asmaa Ibrahim Ahmed
Professor Of Anatomy
And Embryology
Ain Shams University

Endocrine System

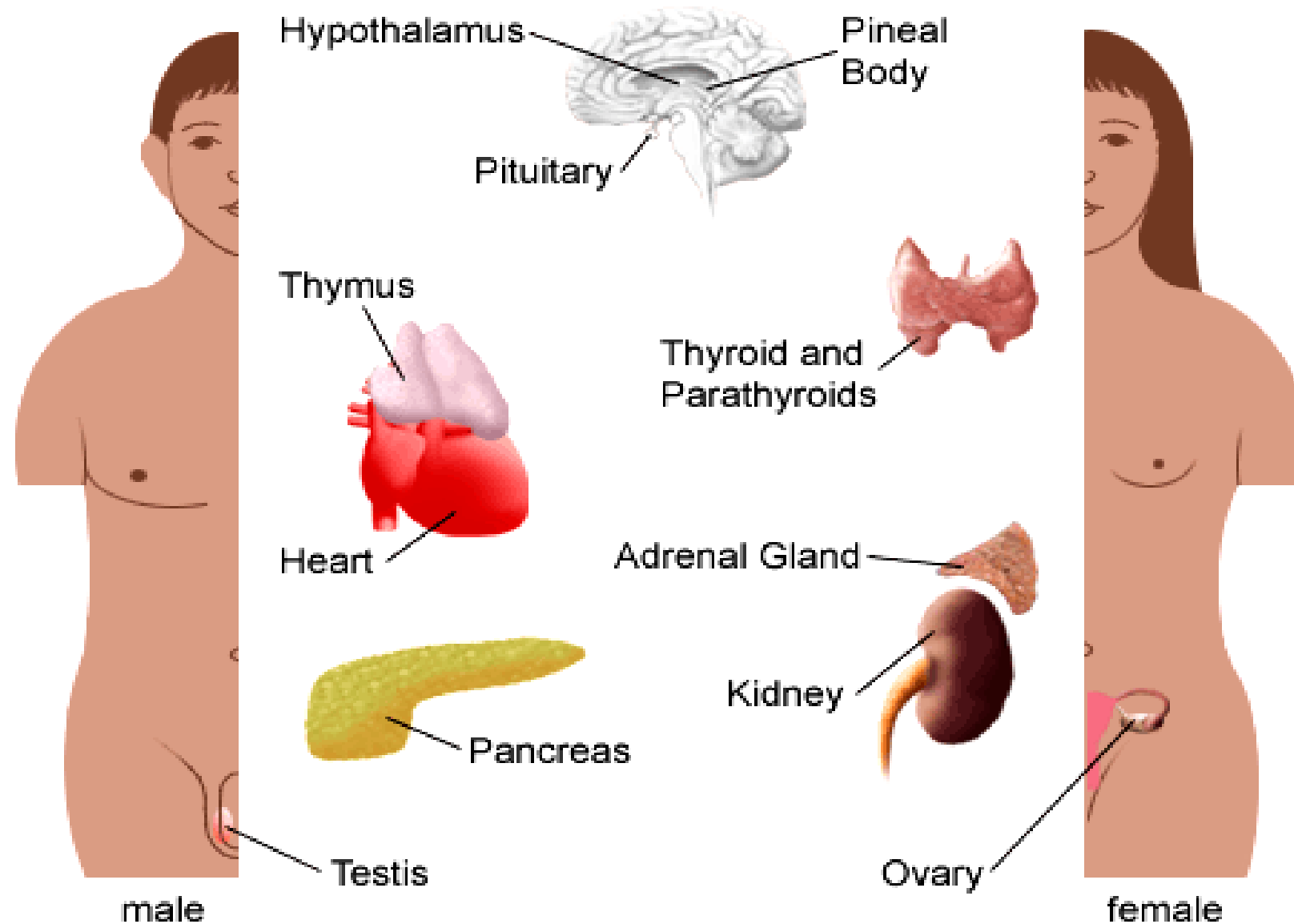


(Fig.45): Endocrine Glands

The Endocrine System

- The **endocrine system** is a system of
- **ductless glands**, each of which secretes a type of **hormone** directly **into the bloodstream** to regulate the body **function**.
- The hormones act on faraway organs

The endocrine system includes:



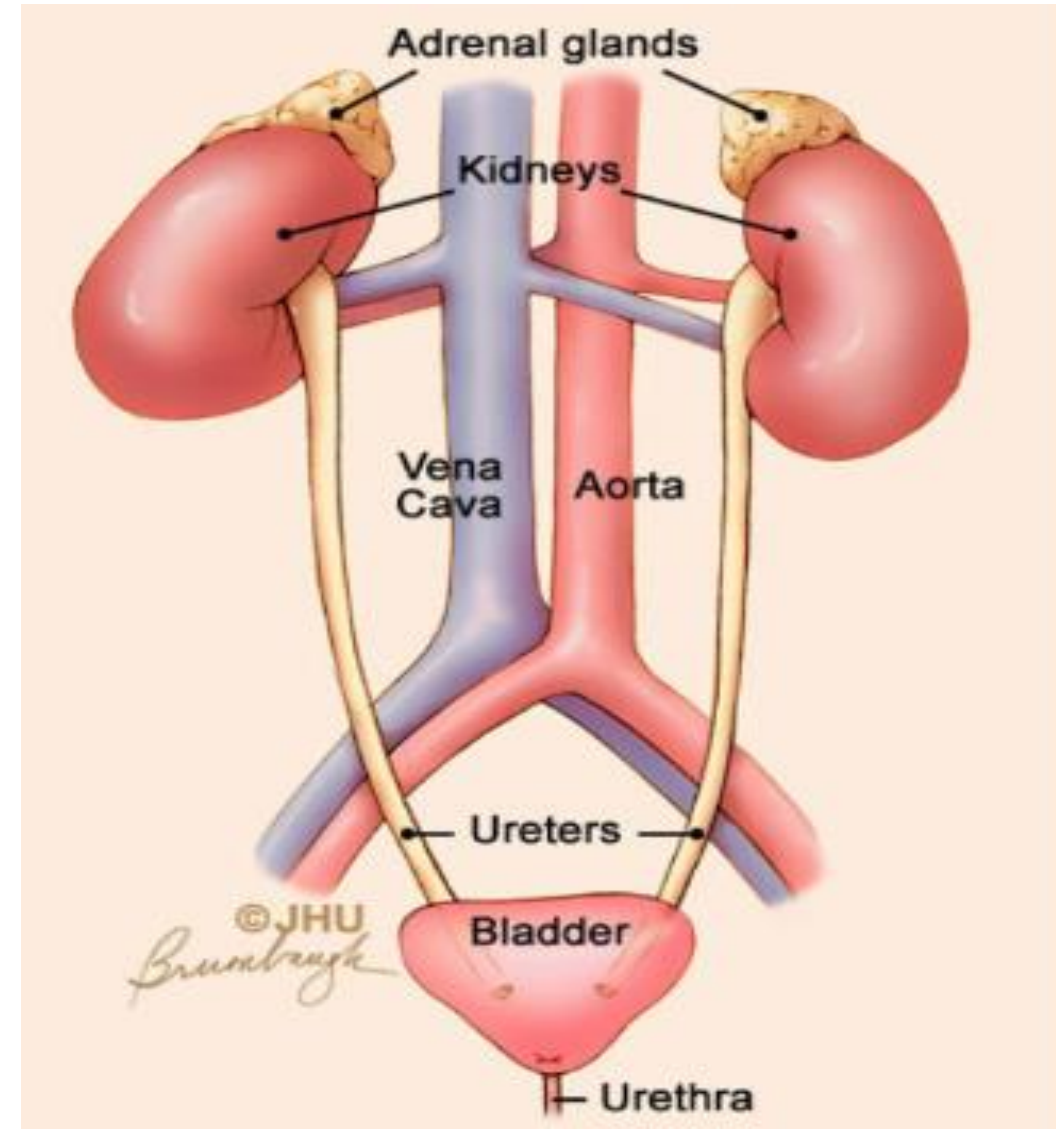
gland	sit	structure	function
Pituitary gland	the base of the brain	anterior lobes Posterior lobe	the master endocrine gland". oxytocin and vasopressin
Thyroid	anterior lower part of the neck.	of two lobes connected together by an isthmus	(T3 and T4)
Parathyroid gland	two pairs in the posterior part of the thyroid gland.		parathyroid hormone
adrenal	Two glands above the kidney.	cortex and medulla	The cortex secretes steroid hormones. The medulla secretes catecholamines.
pancrease	abdomen		secrete insulin.
ovary	pelvis		secrete estrogen and progesterone in females
testes	scrotum		(secrete testosterone) in males

Urinary system

The urinary system formed of

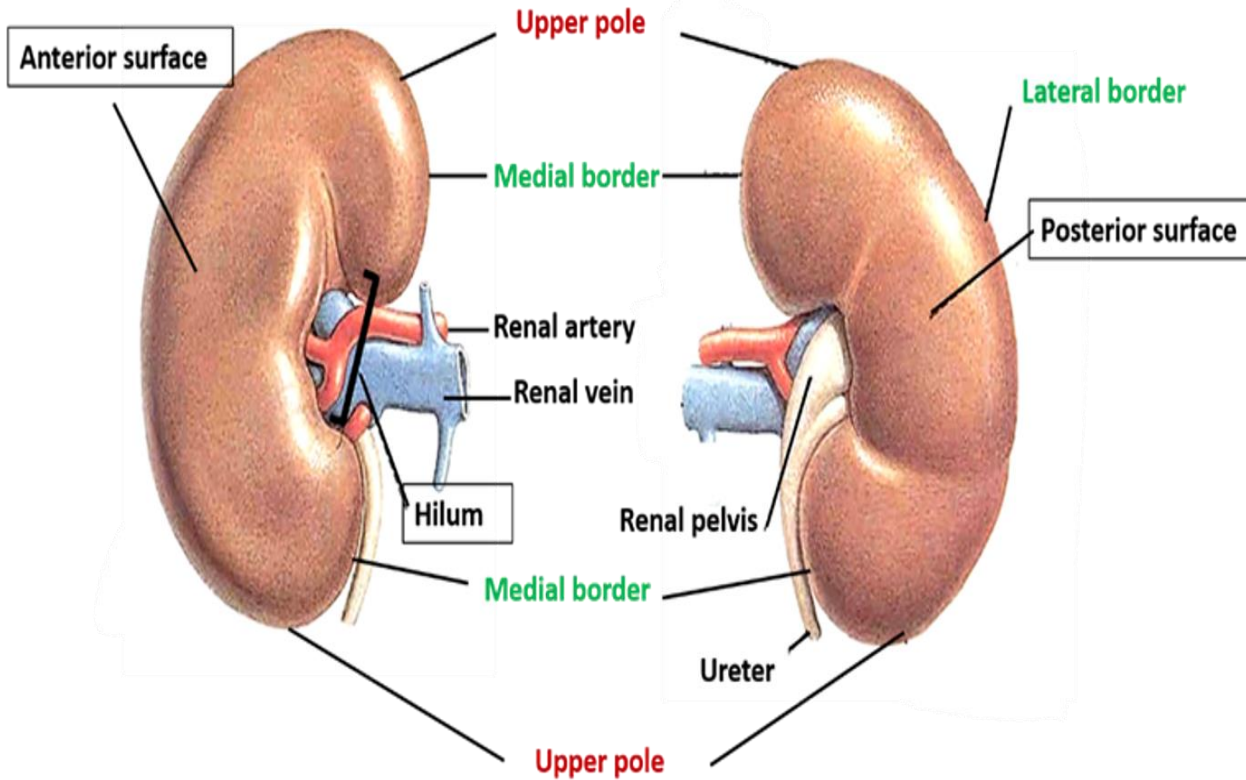
= Pathway of urine:

- 1.the kidneys
2. The ureter
3. the urinary bladder
4. the urethra



1-the kidneys:

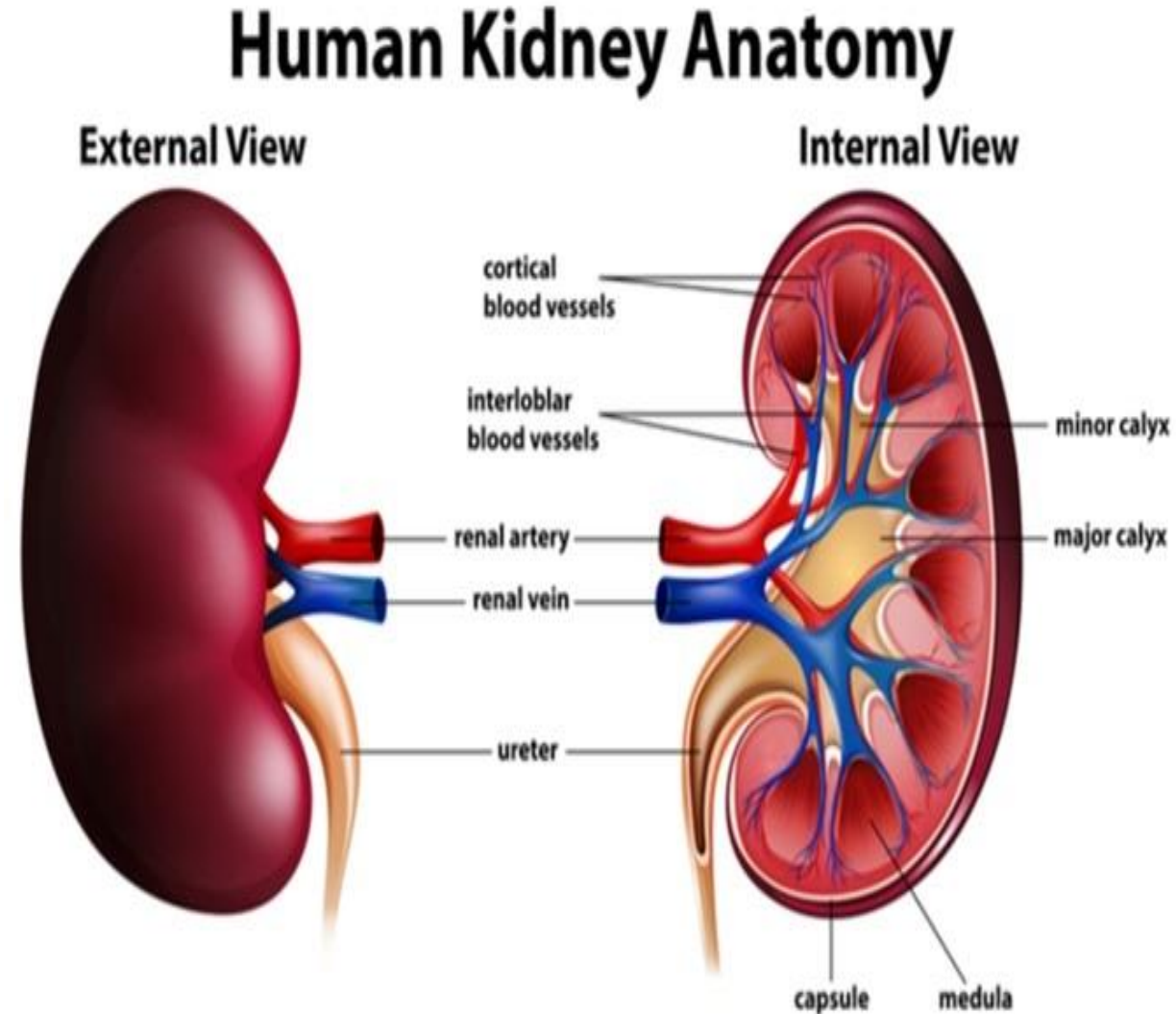
- posterior abdominal wall on **either side of the vertebral column.**
- From 12th thoracic vertebra to 3rd lumbar vertebra.



- Each kidney has **bean-shaped appearance.**
- **2poles:** and superior and inferior poles.
- **2 borders:** medial (concave) and lateral (convex) margins,
- 2 surfaces:** anterior and posterior surfaces. the **medial border** is concave with the superior end of the **ureter.**

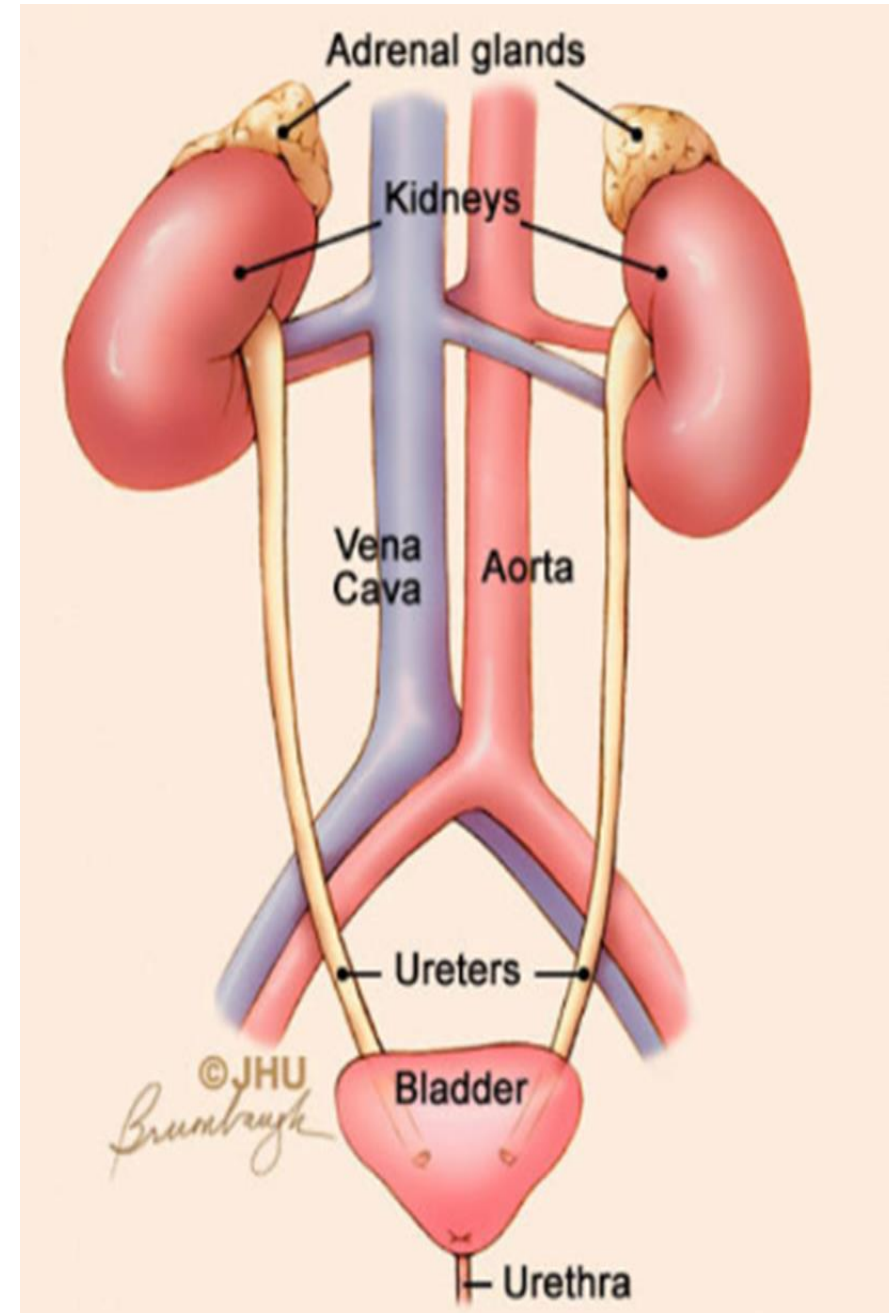
hilum:

- On the **medial border** of each kidney is the hilum
- The hilum transmits, from the front backward (VAU).
 - 1- renal vein
 - 2- renal artery
 - 3- ureters



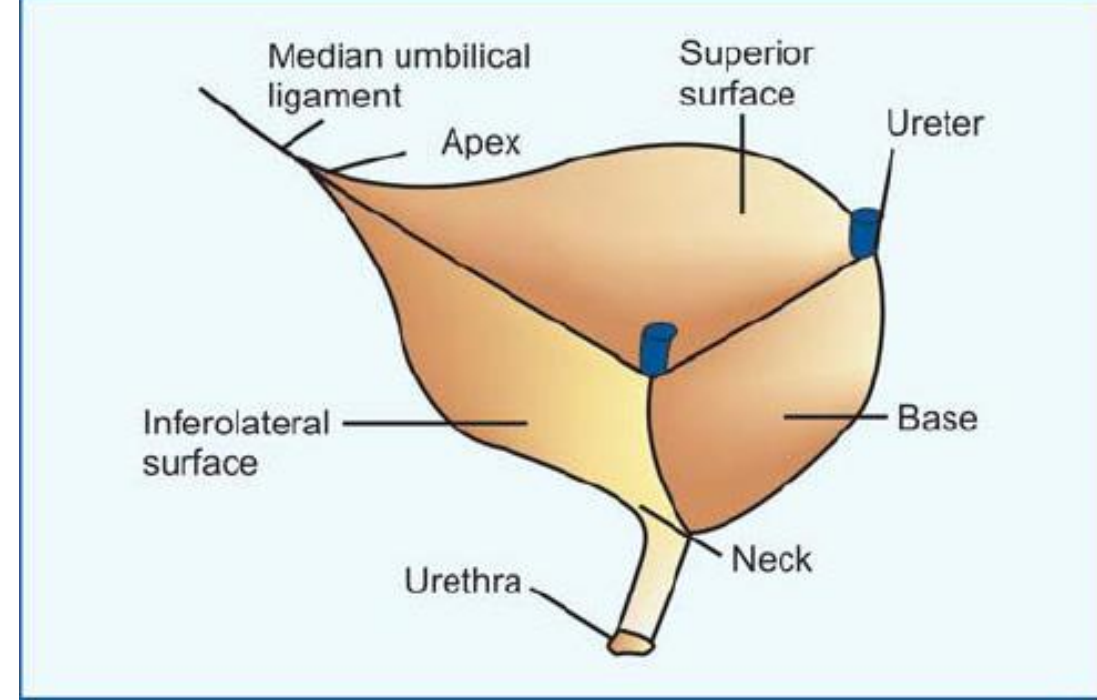
2-Ureter

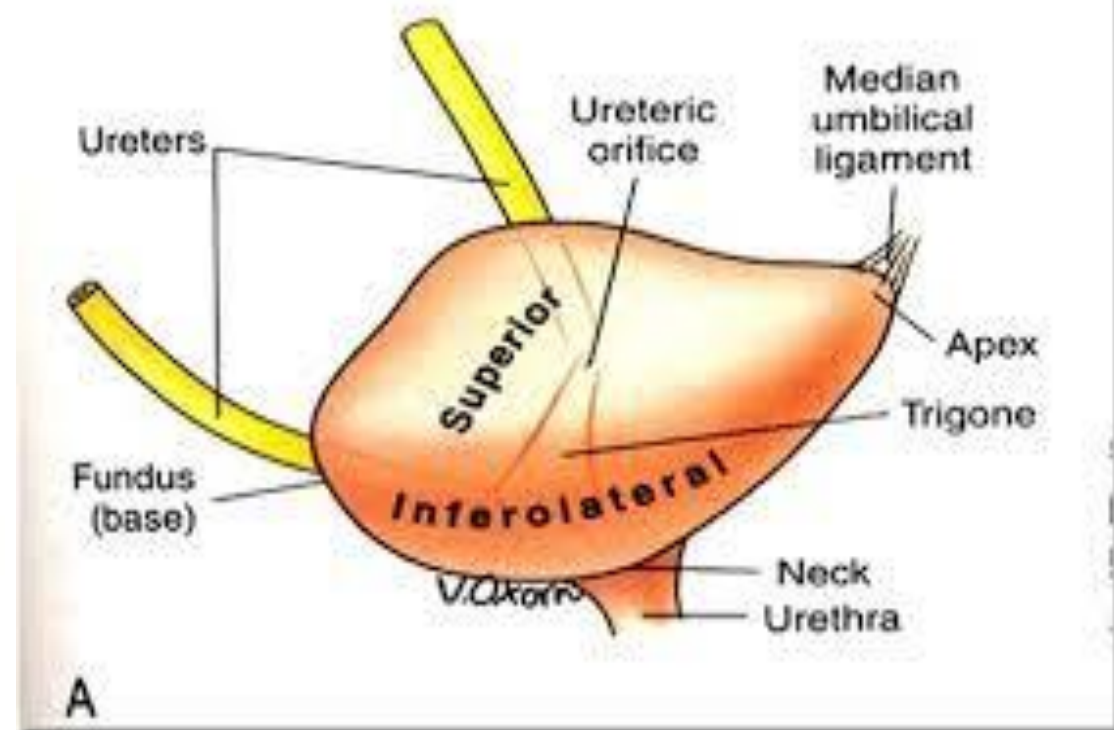
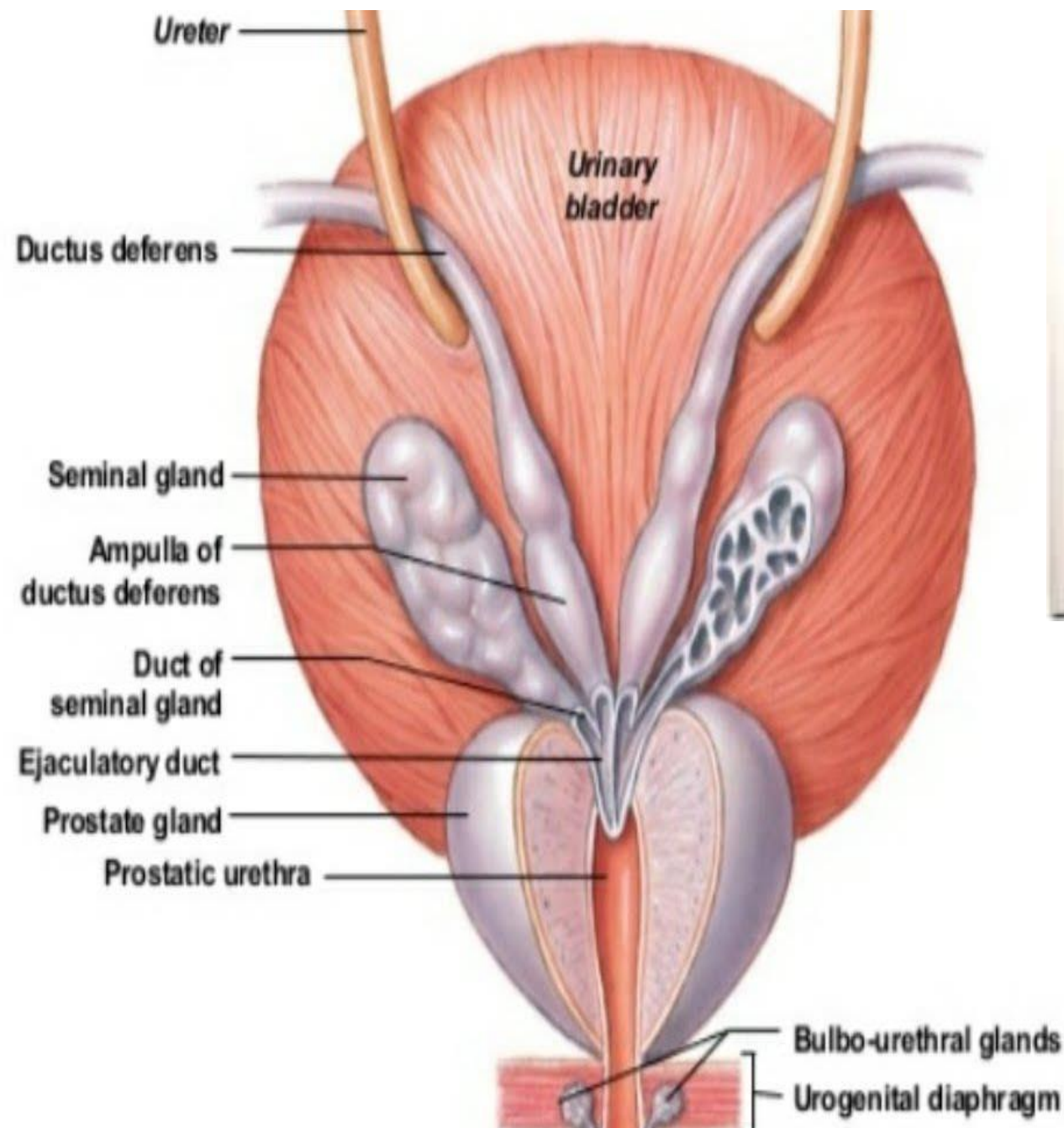
- **Def:** The two ureters **are muscular tubes**,
- **Function:** **transport** the urine **from** the kidney **to** the bladder
- **Sit:** on the posterior abdominal wall **from** the kidneys **to** the urinary bladder.
- **Size:** (25 cm) long
- **structure:** **three constrictions** along its course:
 1. the beginning of the ureter,
 2. In the middle
 3. endit pierces the bladder wall.
- **Clinical:** Ureteric Stones. Renal Colic

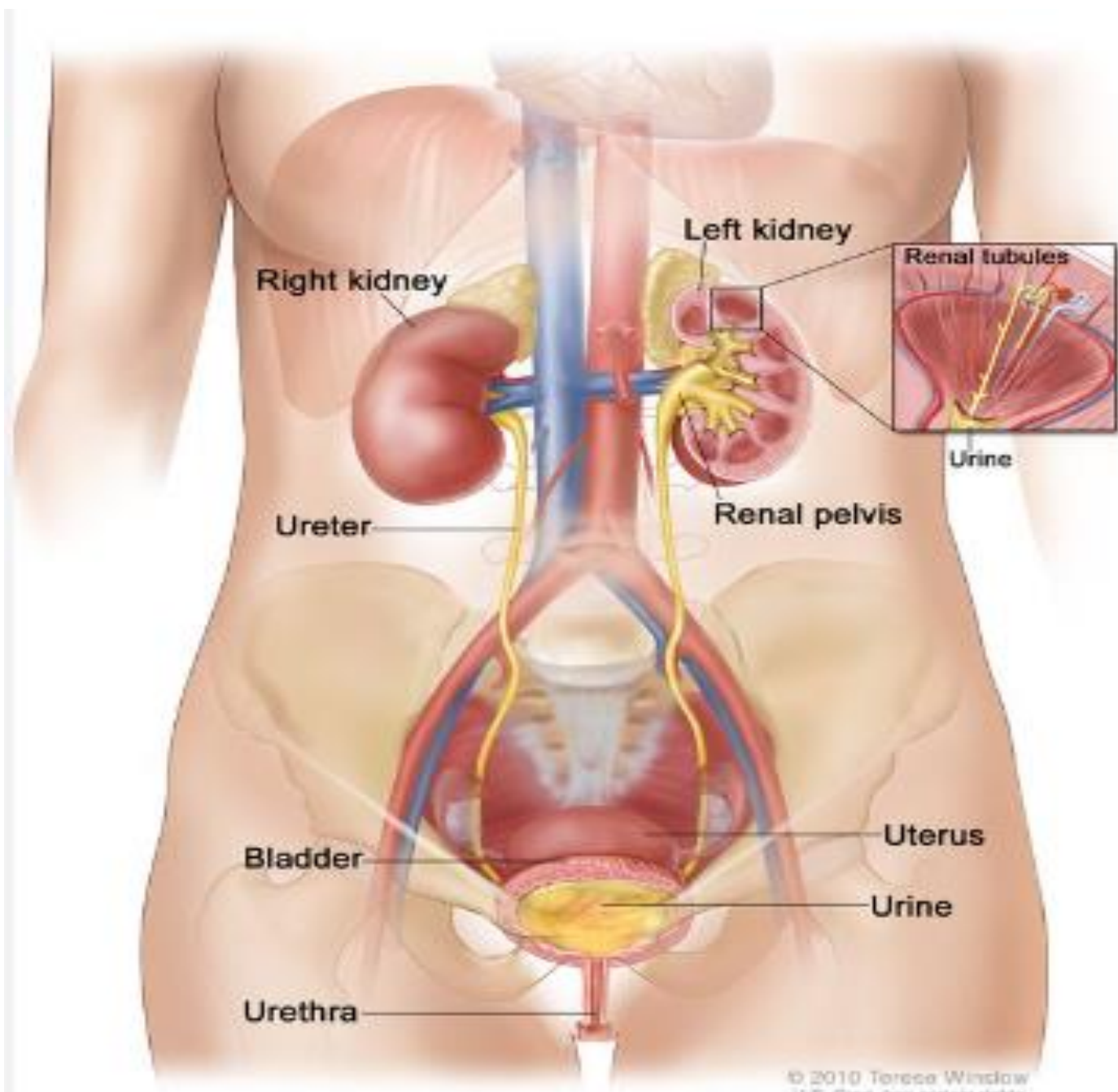


Urinary Bladder

- **function:** It stores urine
- **Sit:** within the pelvis.
- **size:** in adult the capacity 500 mL.
- **shape:** pyramidal
- Structure has a strong muscular wall.
- The empty within the pelvis
- the full bladder, rises up into the abdomen
- having (3) an apex, a base, and a neck
- The apex of the bladder is anteriorly and lies behind the **symphysis pubis**.
- The base, or *posterior surface of the bladder*, Related to the **seminal vesicles**
- *The neck of the bladder lies inferiorly* and rests on the upper surface of the **prostate**.
- The superior angles are joined by the **ureters**
- and the inferior angle gives rise to the **urethra**.



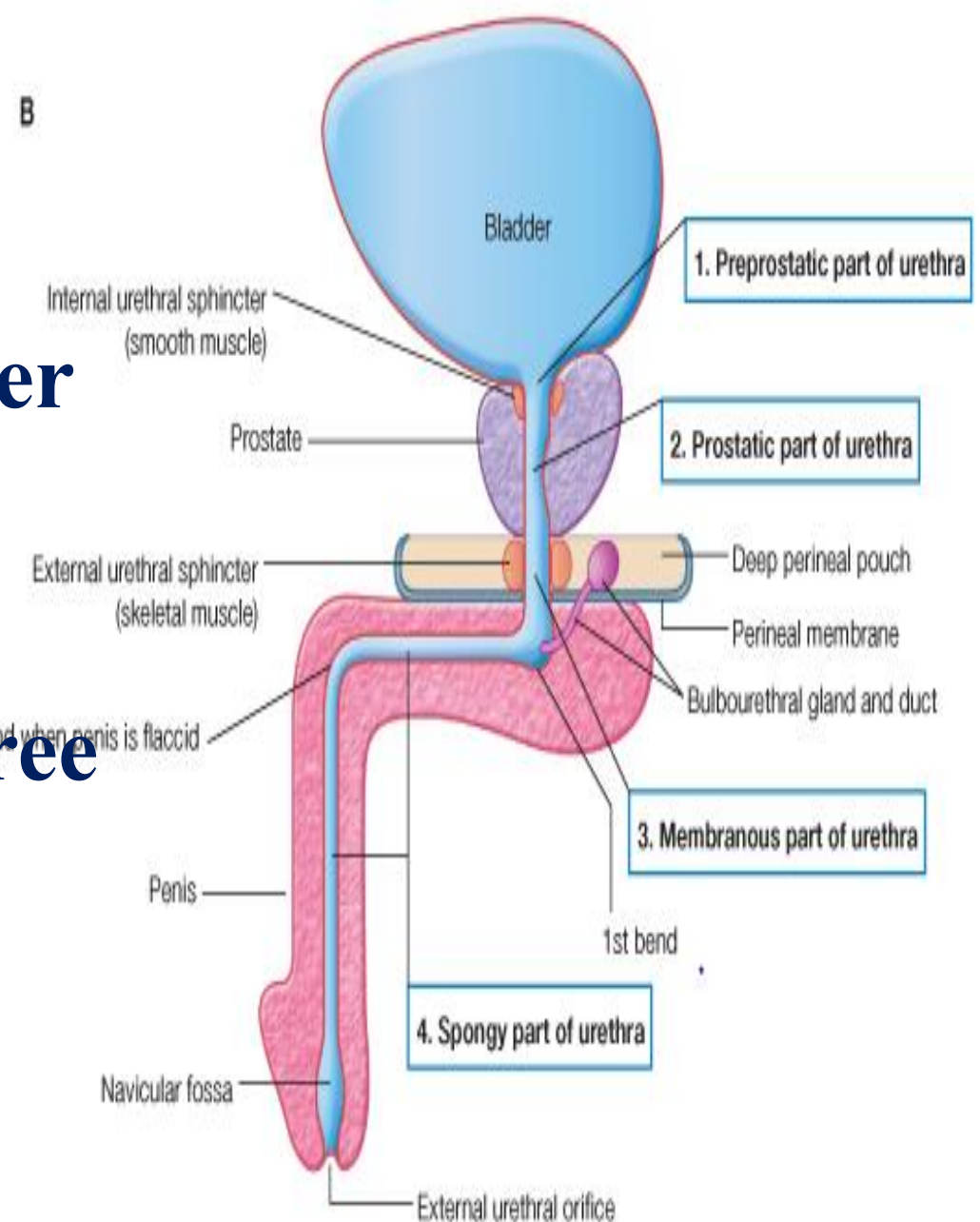




4-Male Urethra

- **Size** : (20 cm) long
- **Sit** : **from the neck of the bladder to the external meatus on the glans penis .**
- **structure** : It is divided **into three parts**:
- **prostatic, membranous, and penile.**

Allow passage of BOTH urine and semen.



5-Female Urethra

- Size: (4 cm).
- Sit: **from the** neck of the **bladder to the** external meatus.
- where it opens into **the vestibule**
- below the clitoris, **It in front of the vagina**

